

2.7 The periodic table — groups 2 and 7

Students will be assessed on their ability to:

1 Properties down group 2

- a explain the trend in the first ionisation energy down group 2
- b recall the reaction of the elements in group 2 with oxygen, chlorine and water
- c recall the reactions of the oxides of group 2 elements with water and dilute acid, and their hydroxides with dilute acid
- d recall the trends in solubility of the hydroxides and sulfates of group 2 elements
- e recall the trends in thermal stability of the nitrates and the carbonates of the elements in groups 1 and 2 and explain these in terms of size and charge of the cations involved
- f recall the characteristic flame colours formed by group 1 and 2 compounds and explain their origin in terms of electron transitions
- g describe and carry out the following:
 - i experiments to study the thermal decomposition of group 1 and 2 nitrates and carbonates
 - ii flame tests on compounds of group 1 and 2
 - iii simple acid-base titrations using a range of indicators, acids and alkalis, to calculate solution concentrations in g dm^{-3} and mol dm^{-3} , e.g. measuring the residual alkali present after skinning fruit with potassium hydroxide
- h demonstrate an understanding of how to minimise the sources of measurement uncertainty in volumetric analysis and estimate the overall uncertainty in the calculated result.

2 Inorganic chemistry of group 7 (limited to chlorine, bromine and iodine)

- a recall the characteristic physical properties of the elements limited to the appearance of solutions of the elements in water and hydrocarbon solvents
- b describe and carry out the following chemical reactions of halogens:
 - i oxidation reactions with metal and non-metallic elements and ions such as iron(II) and iron(III) ions in solution
 - ii disproportionation reactions with cold and hot alkali, e.g. hot potassium hydroxide with iodine to produce potassium iodate(V)
- c carry out an iodine/thiosulfate titration, including calculation of the results and evaluation of the procedures involved, e.g. determination of the purity of potassium iodate(V) by liberation of iodine and titration with standard sodium thiosulfate solution
- d describe and carry out the following reactions:
 - i potassium halides with concentrated sulfuric acid, halogens and silver nitrate solution
 - ii silver halides with sunlight and their solubilities in aqueous ammonia solution
 - iii hydrogen halides with ammonia and with water (to produce acids)
- e make predictions about fluorine and astatine and their compounds based on the trends in the physical and chemical properties of halogens.